

- Maxwellova teorija EM polja povezuje $\vec{E}$ in $\vec{B}$ med seboj in s njihovimi izvori
- Helmholtzov teorem: poljubno vekt. polje je popolnoma določeno, če poznamo njegovo divergenco in rotor
- Ohranjanje naboj (kontinuitetna enačba):

$$\left\{ \begin{array}{l} \oint_V \vec{E} \cdot d\vec{r} = \frac{1}{\epsilon_0} q(t) \\ \oint_V \vec{B} \cdot d\vec{r} = 0 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} \frac{d}{dt} \oint_V \vec{E} \cdot d\vec{r} = \frac{1}{\epsilon_0} \frac{dq(t)}{dt} \\ \frac{d}{dt} \oint_V \vec{B} \cdot d\vec{r} = 0 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} \oint_V \vec{E} \cdot d\vec{r} = \frac{1}{\epsilon_0} q(t) \\ \oint_V \vec{B} \cdot d\vec{r} = 0 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} \oint_V \vec{E} \cdot d\vec{r} = \frac{1}{\epsilon_0} q(t) \\ \oint_V \vec{B} \cdot d\vec{r} = 0 \end{array} \right\}$$

→ Premikalni tok:

- Max. en. v kvazist. približku: $\nabla \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \rightarrow 0 = \nabla \cdot (\nabla \times \vec{B}) = \mu_0 \nabla \cdot \vec{j} \rightarrow$ krši kontinuitetno enačbo

$$\Rightarrow \nabla \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\rightarrow \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \rightarrow 0 = \nabla \cdot (\nabla \times \vec{B}) = \mu_0 \nabla \cdot \vec{j} + \mu_0 \epsilon_0 \nabla \cdot \frac{\partial \vec{E}}{\partial t} = \mu_0 \nabla \cdot \vec{j} + \mu_0 \epsilon_0 \frac{\partial \rho}{\partial t} = 0$$

→ Maxwellove enačbe

$$\left\{ \begin{array}{l} 1) \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \\ 2) \nabla \cdot \vec{B} = 0 \\ 3) \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ 4) \nabla \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \end{array} \right.$$

• enačbe, ki določajo klasičen elektromagnetizem

→ Ohranitveni zakoni

• Max. en. ohranjanje naboja, GK, VK in energijo

$$\left\{ \begin{array}{l} \text{kont. en. za } \vec{E}: \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \\ \text{kont. en. za } \vec{B}: \nabla \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \nabla \cdot \vec{B} = 0 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} -\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} + \vec{B} \frac{\partial \vec{B}}{\partial t} = \vec{E} (\nabla \times \vec{B}) - \vec{B} (\nabla \times \vec{E}) - \mu_0 \vec{j} \cdot \vec{E} / \mu_0 \\ \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon_0 E^2 + \frac{1}{2 \mu_0} B^2 \right) = -\frac{1}{\mu_0} \nabla \cdot (\vec{E} \times \vec{B}) - \vec{j} \cdot \vec{E} \end{array} \right.$$

$$\rightarrow \frac{\partial}{\partial t} \int_V w d^3r + \oint_S \vec{E} \cdot d\vec{r} = -\oint_S \vec{B} \cdot d\vec{r} \rightarrow \frac{\partial}{\partial t} (W_E + W_B) = -\oint_S \vec{B} \cdot d\vec{r} = 0 \Rightarrow \boxed{W_E + W_B = \text{konst.}}$$

• kont. en. za GK

$$\frac{\partial}{\partial t} (\vec{E} \times \vec{B}) = \frac{\partial \vec{E}}{\partial t} \times \vec{B} + \vec{E} \times \frac{\partial \vec{B}}{\partial t} = \frac{1}{\epsilon_0} \left[\frac{1}{\mu_0} \nabla \times \vec{B} \times \vec{B} - \frac{\mu_0}{\epsilon_0} \vec{j} \times \vec{B} - \vec{E} \times \nabla \times \vec{E} \right] = -\epsilon_0 \left[\nabla \cdot (\vec{E} \otimes \vec{E}) + \frac{1}{2} \nabla E^2 + \frac{c^2}{2} \nabla B^2 - c^2 \nabla (\vec{B} \otimes \vec{B}) \right] - (\vec{E} \cdot \vec{j} + \vec{j} \cdot \vec{E})$$

$$\rightarrow \frac{\partial}{\partial t} \epsilon_0 (\vec{E} \times \vec{B}) - \nabla \cdot \left[\epsilon_0 (\vec{E} \otimes \vec{E}) + \frac{1}{2} \epsilon_0 \nabla E^2 + \frac{1}{\mu_0} (\vec{B} \otimes \vec{B}) - \frac{1}{2} \frac{1}{\mu_0} \nabla B^2 \right] + (\vec{E} \cdot \vec{j} + \vec{j} \cdot \vec{E}) = 0 \Rightarrow \left\{ \begin{array}{l} \frac{\partial g_i}{\partial t} - \frac{\partial T_{ik}}{\partial x_k} + f_i = 0 \\ \frac{\partial g_i}{\partial t} - \frac{\partial T_{ik}}{\partial x_k} + f_i = 0 \end{array} \right.$$

$$\rightarrow \frac{d}{dt} \int_V g_i d^3r + \oint_S T_{ik} n_k dS = \oint_S T_{ik} n_k dS = 0 \Rightarrow \boxed{\vec{G}_E + \vec{G}_B = \text{konst.}}$$

• Ohranjanje VK

$$\frac{\partial}{\partial t} g_i - \frac{\partial T_{ik}}{\partial x_k} + f_i = 0 \quad / \quad x_i, \text{ neposredno } \frac{\partial x_i}{\partial x_j} = \delta_{ij}$$

$$\frac{\partial}{\partial t} x_j g_i - \frac{\partial x_j T_{ik}}{\partial x_k} + T_{ji} + x_j f_i = 0 \quad / \quad \epsilon_{ijk}, \text{ antisim. tenzor, } \epsilon_{iji} T_{ji} = 0$$

$$\rightarrow \frac{\partial}{\partial t} \epsilon_{ijk} x_j g_i - \frac{\partial \epsilon_{ijk} x_j T_{ik}}{\partial x_k} + \epsilon_{iji} x_j f_i = 0 \quad \text{kont. en. za VK} \rightarrow \frac{\partial}{\partial t} \epsilon_{ik} - \frac{\partial \epsilon_{ij} x_j T_{ik}}{\partial x_k} + m_i = 0 \quad / \quad \int d^3r \rightarrow \frac{\partial}{\partial t} \int_V g_i d^3r = \int_V \epsilon_{ijk} x_j T_{ik} n_k dS - \int_V m_i d^3r$$

$$\Gamma_i^E = \int_V \epsilon_{ijk} x_j g_i d^3r = \int_V (\vec{r} \times \vec{g})_i d^3r \quad \text{Celotna VK elektromag. polja}$$

$$\vec{\Gamma}^E = \int_V \vec{r} \times \vec{g} d^3r$$

$$M_i^m = \int_V \epsilon_{ijk} x_j f_i d^3r = \int_V (\vec{r} \times \vec{f})_i d^3r \quad \text{Mehanski moment na snov} \Rightarrow \frac{d}{dt} (\vec{\Gamma}^E + \vec{\Gamma}^m) = \int_V M_{ik} n_k dS = 0 \Rightarrow \boxed{\vec{\Gamma}^E + \vec{\Gamma}^m = \text{konst.}}$$

$$\vec{M}^m = \int_V \vec{r} \times \vec{f} d^3r = \frac{d}{dt} \vec{\Gamma}^m$$

$$M_{ik} = \epsilon_{ijk} x_j T_{ik} \quad \text{Tenzor gostote toka VK}$$